Homework 4 Math 541 B

Exercise 1. Write down the generalized likelihood ratio estimate for the following alpha particle data, as we did in class for a slightly different data set. The corresponding test treats individual counts of alpha particles as independent Poisson random variables, versus the alternative that the probability of a count appearing in each box of data is a sequence of nonnegative numbers that sum to one. In doing so, compute a maximum likelihood estimate using a computer.

m	$0,1,2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	≥ 17
Intervals	16	26	58	102	125	146	163	164	120	100	72	54	20	12	10	4

Plot the maximum likelihood function for the Poisson statistic, that is, the denominator of the generalized likelihood ratio statistic,

$$\sup_{\theta \in \Theta_0} f_\theta(X),$$

as a function of λ. Finally, compute the value s of Pearson's chi-squared statistic S, and compute $P(S \geq s)$. Does this probability give confidence that the null hypothesis is true?

Solution: There are $N = 1192$ intervals. Treat the first and last boxes as the categories $\{0,1,2\}$ and $\{17,18,\ldots\}$, respectively. For a Poisson random variable with mean λ, the probabilities of the sixteen categories are

$$p_1(\lambda) = e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right), \qquad p_j(\lambda) = e^{-\lambda}\frac{\lambda^{j+1}}{(j+1)!}, \quad 2 \leq j \leq 15,$$

and

$$p_{16}(\lambda) = 1 - \sum_{r=0}^{16} e^{-\lambda}\frac{\lambda^r}{r!}.$$

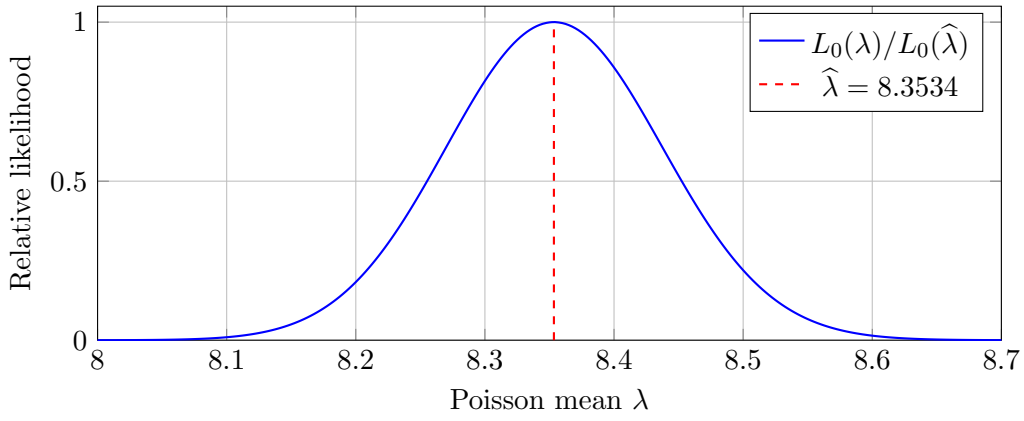
Here p_2 is the probability of count 3, and p_{15} is the probability of count 16. If N_j denotes the observed frequency in category j, the null likelihood, up to the multinomial coefficient common to both models, is

$$L_0(\lambda) = \prod_{j=1}^{16} p_j(\lambda)^{N_j}, \qquad \ell_0(\lambda) = \sum_{j=1}^{16} N_j \log p_j(\lambda).$$

Numerical maximization gives

$$\widehat{\lambda} = 8.3534369, \qquad \ell_0(\widehat{\lambda}) = -2938.019714.$$

The denominator of the generalized likelihood ratio is $L_0(\lambda)$; the following plot shows it divided by its maximum, since the unscaled likelihood is extremely small.



Under the unrestricted alternative, the maximum likelihood estimates of the sixteen category probabilities are $\widehat{q}_j = N_j/N$. The maximized log likelihood, again omitting the common multinomial coefficient, is

$$\ell_1 = \sum_{j=1}^{16} N_j \log(N_j/N) = -2932.622622.$$

Thus the generalized likelihood ratio and its deviance form are

$$\Lambda = \frac{\sup_{H_0} L_0}{\sup_{H_1} L_1} = e^{\ell_0(\widehat{\lambda})-\ell_1} \approx 0.00453, \qquad -2\log\Lambda \approx 10.7942.$$

At $\widehat{\lambda}$, the expected category counts are approximately

$$(12.424, 27.282, 56.974, 95.185, 132.521, 158.143, 165.130, 153.267,$$
$$128.031, 97.227, 67.682, 43.490, 25.950, 14.451, 7.545, 6.700).$$

Pearson's statistic is

$$S = \sum_{j=1}^{16} \frac{(N_j - Np_j(\widehat{\lambda}))^2}{Np_j(\widehat{\lambda})} = 10.7989.$$

There are $16 - 1 - 1 = 14$ degrees of freedom, accounting for the probability constraint and the estimated parameter. Consequently,

$$P(\chi^2_{14} \geq 10.7989) \approx 0.7018.$$

This p-value says that the observed discrepancy is not unusual under the fitted Poisson model. It is not the probability that the null hypothesis is true, and it does not prove the null hypothesis.

Exercise 2. Let $X_1, \ldots, X_n$ be i.i.d. random variables and let $0 < \alpha < 1/2$. Define the α-trimmed sample mean by

$$X_n^{(\alpha)} = \frac{1}{n - 2\lfloor n\alpha \rfloor} \sum_{i=\lfloor n\alpha \rfloor + 1}^{n-\lfloor n\alpha \rfloor} X_{(i)}.$$

For any $w = (w_1, \ldots, w_n) \in \{1, \ldots, n\}^n$, define the Winsorized sample mean by

$$X_n^{(w)} = \frac{1}{n} \sum_{i=1}^{n} X_{(w_i)}.$$

Show that the jackknife estimate of $X_n^{(\alpha)}$ can be written as

$$\frac{1}{1-2\alpha}\left(X_n^{(w)} - 2\alpha X_n^{(\alpha)}\right) + O(n^{-1})$$

for some vector w, plus another quantity that is $O(1/n)$, assuming the random variables are bounded. Show that the jackknife variance estimate of $X_n^{(\alpha)}$ is asymptotically equal to

$$\frac{1}{n(n-1)(1-2\alpha)^2} \sum_{i=1}^{n} \left(X_{(w_i)} - X_n^{(w)}\right)^2$$

for some w.

Solution: Write $k = \lfloor n\alpha \rfloor$, $m = n - 2k$, and

$$T = X_n^{(\alpha)} = \frac{1}{m} \sum_{i=k+1}^{n-k} X_{(i)}.$$

First use the delete-one convention that keeps the trim count equal to k. Define the Winsorized values

$$W_i = X_{(w_i)}, \qquad w_i = \begin{cases} k+1, & i \le k, \\ i, & k < i \le n-k, \\ n-k, & i > n-k. \end{cases} \qquad \overline{W} = \frac{1}{n} \sum_{i=1}^{n} W_i.$$

If observation i is deleted, the sum of the remaining central observations is $mT - W_i$. Hence

$$T_{(-i)} = \frac{mT - W_i}{m-1}, \qquad \overline{T}_{(-)} = \frac{mT - \overline{W}}{m-1}.$$

The jackknife point estimate is therefore

$$\begin{aligned} T_J &= nT - (n-1)\overline{T}_{(-)} \\ &= \frac{n-1}{m-1}\overline{W} - \frac{2k}{m-1}T. \end{aligned}$$

Since $k = n\alpha + O(1)$ and $m = n(1-2\alpha) + O(1)$, the boundedness assumption gives

$$T_J = \frac{\overline{W} - 2\alpha T}{1-2\alpha} + O(n^{-1}).$$

The jackknife variance estimate is

$$\begin{aligned} \widehat{\mathrm{Var}}_J(T) &= \frac{n-1}{n} \sum_{i=1}^{n} (T_{(-i)} - \overline{T}_{(-)})^2 \\ &= \frac{n-1}{n(m-1)^2} \sum_{i=1}^{n} (W_i - \overline{W})^2 \\ &= \frac{1 + O(n^{-1})}{n(n-1)(1-2\alpha)^2} \sum_{i=1}^{n} (W_i - \overline{W})^2. \end{aligned}$$

The final equality uses $m = n(1-2\alpha) + O(1)$. Since the observations are bounded, the sum is $O(n)$, so the absolute difference between the last expression and the requested approximation is $O(n^{-2})$.

There is a floor-function detail in the usual delete-one jackknife. If the trim count is recomputed on the $n-1$ observations, then $k_- = \lfloor (n-1)\alpha \rfloor$ can equal $k-1$. In that case define $B = X_{(k)} + X_{(n-k+1)}$ and Winsorize at ranks k and $n-k+1$, obtaining values W_i'. The leave-one-out trimmed means are then

$$T_{(-i)} = \frac{mT + B - W_i'}{m+1}.$$

The variance calculation still gives the stated first-order variance formula with W_i'. The point-estimator calculation above assumes the trim count is kept fixed; with a recomputed floor, the claimed point-estimator expansion requires an additional convention or assumption to handle the case $k_- = k - 1$.

Exercise 3. Let X_1, X_2, X_3 be i.i.d. continuous random variables such that X_1 has probability density function f_θ for $\theta \in \Theta$. Let W_1, W_2, W_3 be a bootstrap sample from X_1, X_2, X_3. Let Y denote the sample median of X_1, X_2, X_3, that is, the middle value among the three observations, which is unique with probability one since the random variables are continuous. Describe the distribution of $(W_{(1)}, W_{(2)}, W_{(3)})$, the bootstrap estimator of Y, and the bootstrap estimator of the variance of Y.

Solution: Condition on the observed values and write them in increasing order as $x_{(1)} < x_{(2)} < x_{(3)}$. Each bootstrap draw selects one of these values with probability $1/3$. Thus the ordered bootstrap triple has the following distribution:

- $(x_{(i)}, x_{(i)}, x_{(i)})$ has probability $1/27$ for each $i \in \{1, 2, 3\}$;
- each of the six triples with two copies of one value and one copy of a different value has probability $1/9$;
- $(x_{(1)}, x_{(2)}, x_{(3)})$ has probability $2/9$.

The bootstrap median $W_{(2)}$ has distribution

$$P_*(W_{(2)} = x_{(1)}) = \frac{7}{27}, \qquad P_*(W_{(2)} = x_{(2)}) = \frac{13}{27}, \qquad P_*(W_{(2)} = x_{(3)}) = \frac{7}{27}.$$

Therefore, the bootstrap estimator of the median is

$$\widehat{Y}_{\text{boot}} = E_*[W_{(2)}] = \frac{7}{27}x_{(1)} + \frac{13}{27}x_{(2)} + \frac{7}{27}x_{(3)}.$$

The bootstrap variance estimate is

$$\begin{aligned}\widehat{\text{Var}}_{\text{boot}}(Y) &= \sum_{j=1}^{3} p_j (x_{(j)} - \widehat{Y}_{\text{boot}})^2 \\ &= \frac{91(x_{(1)} - x_{(2)})^2 + 49(x_{(1)} - x_{(3)})^2 + 91(x_{(2)} - x_{(3)})^2}{729},\end{aligned}$$

where $(p_1, p_2, p_3) = (7/27, 13/27, 7/27)$. These are the exact conditional bootstrap quantities; finitely many bootstrap repetitions approximate them by Monte Carlo.

Exercise 4. Let $\mu \in \mathbb{R}$ and let $0 < \sigma < \infty$. Let $X_1, \ldots, X_n$ be i.i.d. real-valued random variables each with mean μ and variance σ^2. Let $h : \mathbb{R} \to \mathbb{R}$ be a function such that h' exists and is continuous. Set

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad Y_n = h(\overline{X}_n),$$

Show that the jackknife estimate of the variance of Y_n converges almost surely to the same variance estimate obtained by applying the Delta Method to Y_n.

Solution: Let $\overline{X}_n = n^{-1}\sum_i X_i$ and let

$$X_{(-i)} = \frac{n\overline{X}_n - X_i}{n-1}, \qquad Y_{(-i)} = h(X_{(-i)}), \qquad \overline{Y}_{(-)} = \frac{1}{n}\sum_{i=1}^{n} Y_{(-i)}.$$

The jackknife variance estimator is

$$\widehat{V}_J = \frac{n-1}{n}\sum_{i=1}^{n}(Y_{(-i)} - \overline{Y}_{(-)})^2.$$

Put $\delta_i = X_{(-i)} - \overline{X}_n = -(X_i - \overline{X}_n)/(n-1)$. Then $\sum_i \delta_i = 0$. By the strong law, $\overline{X}_n \to \mu$ and the sample variance

$$s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$$

converges almost surely to σ^2. Also $\max_{i\le n}|X_i|/n \to 0$ almost surely, so all $X_{(-i)}$ approach μ uniformly over i. Continuity of h' therefore gives a sequence $\varepsilon_n \to 0$ almost surely for which Taylor's formula yields

$$Y_{(-i)} = h(\overline{X}_n) + h'(\overline{X}_n)\delta_i + r_{ni}, \qquad |r_{ni}| \le \varepsilon_n|\delta_i|.$$

After centering over i, the constant term vanishes. Since

$$\sum_{i=1}^{n}\delta_i^2 = \frac{s_n^2}{n-1},$$

the Taylor remainder contributes $o(n^{-1})$ almost surely to the jackknife variance. Consequently,

$$\widehat{V}_J = \frac{h'(\overline{X}_n)^2 s_n^2}{n} + o(n^{-1}) \quad \text{almost surely.}$$

The Delta Method gives asymptotic variance $h'(\mu)^2\sigma^2/n$. Using $h'(\overline{X}_n) \to h'(\mu)$ and $s_n^2 \to \sigma^2$ almost surely, we obtain

$$n\widehat{V}_J \longrightarrow h'(\mu)^2\sigma^2 \quad \text{almost surely,}$$

which is the Delta Method variance to first order.

Exercise 5. Suppose $X_1, \ldots, X_n$ is a random sample from a Gaussian random variable X with unknown mean $\mu_X \in \mathbb{R}$ and unknown variance $\sigma^2 > 0$. Suppose $Y_1, \ldots, Y_m$ is a random sample from a Gaussian random variable Y with unknown mean $\mu_Y \in \mathbb{R}$ and unknown variance $\sigma^2 > 0$. Assume that the two samples are independent and that $n + m > 2$. Define

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n}X_i, \qquad \overline{Y} = \frac{1}{m}\sum_{i=1}^{m}Y_i,$$

$$S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2, \qquad S_Y^2 = \frac{1}{m-1}\sum_{i=1}^{m}(Y_i - \overline{Y})^2,$$

and the pooled variance

$$S^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}, \qquad n+m>2.$$

Show that

$$\frac{\overline{X} - \overline{Y} - \mu_X + \mu_Y}{S\sqrt{1/n + 1/m}}$$

has a Student t distribution with $n+m-2$ degrees of freedom. Deduce the following coverage probability for the difference of the means, where $p = n+m-2$:

$$P\left(\overline{X}-\overline{Y}-tS\sqrt{\frac{1}{n}+\frac{1}{m}} < \mu_X-\mu_Y < \overline{X}-\overline{Y}+tS\sqrt{\frac{1}{n}+\frac{1}{m}}\right)$$
$$= \frac{\Gamma((p+1)/2)}{\sqrt{p\pi}\,\Gamma(p/2)}\int_{-t}^{t}\left(1+\frac{s^2}{p}\right)^{-(p+1)/2} ds.$$

Solution: Let $p = n+m-2$. Since the two samples are independent and Gaussian,

$$Z = \frac{\overline{X}-\overline{Y}-(\mu_X-\mu_Y)}{\sigma\sqrt{1/n+1/m}} \sim N(0,1).$$

The sample means are independent of the within-sample residual sums of squares. Also,

$$\frac{(n-1)S_X^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \frac{(m-1)S_Y^2}{\sigma^2} \sim \chi^2_{m-1}.$$

The two terms are independent and are independent of Z. Their sum gives

$$U = \frac{(n-1)S_X^2+(m-1)S_Y^2}{\sigma^2} = \frac{pS^2}{\sigma^2} \sim \chi^2_p, \qquad U \text{ independent of } Z.$$

It follows from the normal-over-independent-chi-squared definition of the Student distribution that

$$\frac{\overline{X}-\overline{Y}-\mu_X+\mu_Y}{S\sqrt{1/n+1/m}} \sim t_p.$$

For $t > 0$, the stated event is equivalent to the event that this pivot lies in $(-t,t)$. The density of a t_p random variable is

$$f_p(s) = \frac{\Gamma((p+1)/2)}{\sqrt{p\pi}\,\Gamma(p/2)}\left(1+\frac{s^2}{p}\right)^{-(p+1)/2}.$$

Therefore, integrating the t_p density over $(-t,t)$ gives the stated coverage probability. For a confidence level $1-\gamma$, choose t to be the $1-\gamma/2$ quantile of t_p. The resulting confidence interval is

$$\left(\overline{X}-\overline{Y}\right) \pm t_{p,1-\gamma/2}S\sqrt{\frac{1}{n}+\frac{1}{m}}.$$

If either sample size is one, its displayed sample variance is undefined; the result extends by interpreting that group's residual sum of squares as zero. Otherwise both sample variances in the question require $n, m \geq 2$.